



CVD – Black

CVD for Dressing

Our dressing logs are created using DC arc Plasma Jet technology.

This process produces a much higher thermal conductivity product than those produced by other techniques and our high growth rates does not compromise the quality of our products. For high speed dressing operations this is critical.

High fracture strength and thermal conductivity combined with high thermal oxidation resistance and high fracture toughness makes it the product of choice for high performance dressing tools.

Applications

CVD diamond dresser blanks delivered by EID is suitable for all dresser types - single point, multipoint, blade, rotary, roller etc. The dresser size is determined by the application and all dimensions can be delivered on request.

Our blanks and laser cut CVD allows the optimum setting environment as demanded for all types of CVD tooling industrial applications, using traditional non-ferrous metal sintering.

CVD diamond dressers can also be brazed onto a dresser body using an active brazed alloy in a non-oxidizing environment.

Benefits

- Excellent thermal stability
- High wear resistance to chipping and fracture
- Performance independent of dressing orientation
- Diamond with no binder phase
- Uniform wear resistance over the entire length of the dresser

Availability

Our CVD diamond dresser is supplied as rectangular logs in various lengths and the cross-section dimensions. CVD diamond dresser products with almost all non-standard dimensions are available on request.

It is available in our High-grade HQ or SQ standard grade.

Cutting tools

Due to the extraordinary high concentration of atomic hydrogen produced by the arc discharge, cutting tool blanks fabricated by DC arc Plasma Jet offer substantially higher thermal conductivity than those by other techniques even at a relatively high growth rate. Offering apparently favorable conditions for heavy load high speed cutting operations. Very low friction coefficient, higher fracture strength and fracture toughness.

Our synthesis process and a state-of-the-art quality control process enables us to provide a comprehensive range of CVD diamond for the dressing and cutting tools industry.

Advantages

- Excellent wear resistance for a longer cutting duration
- Low friction coefficient for reducing the processing machining heat
- Perfect thermal conductivity for decreasing the temperature at the cutting edge
- High fracture strength and toughness for spicatto cutting
- Better dimensional and thermal stability for achieving good cutting quality





Applications

CVD diamond cutting tools are widely used for cutting (machining) non-ferrous materials, ceramics, abrasive composites, high silicon-content aluminum alloy, graphite, and other difficult machining materials.

Thermal Management

One of the remarkable properties of diamond is its unsurpassed thermal conductivity, which is five times higher than that of copper. As we all know, heat is conducted by electrons in metal, however, lattice vibrations are the key factor for the high thermal conductivity of diamond.

Due to the extraordinary high concentration of atomic hydrogen produced by the very high temperatures of the arc discharging, diamond films with high thermal conductivity can be easily obtained even at relatively high growth rate by DC Arc Plasma Jet. The thermal conductivity of our optical quality CVD diamond made by our company through DC Arc Plasma Jet approaches that of the natural type IIa diamond, 2000 W/mK. Thermal conductivity can be tailored within the range of 800-2000 W/mK, and thus can be subdivided into different quality grades to meet the different needs of our customers in different application fields. At present, CVD diamond made by our company is used as mounts for high-power integrated circuits, laser diodes, GaN on Diamond (GOD) and heat spreaders for satellite based electronic systems.

	Standard Grade SQ	High Grade HQ
Process	Dc Arc Plasma	Dc Arc Plasma
Chemical composition (%)	100% Carbon	100% Carbon
density (g/cm ³)	3.52	3.52
Poissons ratio	0.1	0.1
Youngs modulus (Gpa)	1000-1100	1000-1100
Dielectric constant	5.7	5.7
Electric resistivity (ohm/cm)	>10exp14	>10exp14
Dielectric strength (v/cm)	1x10exp7	1x10exp7
Loss tangent	<4x10-4	<4x10-4
Thermal conductivity (W/mk)	>10W/cm.K	>10W/cm.K
Tensile strength (kg/mm ²) 0.50 mm thick growth surface tension	<350	>550
Vickers hardness (kg/mm ²)	7000-10000	7000-10000
Compressive strength (Gpa)	<1100GPa	<1100GPa
CTE	1.0x10exp-6/k @ 300k	1.0x10exp-6/k @ 300k
Thermal stability	800 degrees C	800 degrees C
Wear resistant (abrasion ratio)	100-200,000	>400,000
Breaking resistance for 0.50 mm thickness (Mpa)	see tensile strength	see tensile strength
Polishing		
Surface finish sub side	<30 nm	<30 nm
Surface finish growth side	<100 nm	<100 nm
Thickness tolerance	plus minus 0.05 mm	plus minus 0.05 mm